APS360 Final Report: AI-Generated Music Detection

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https://github.com/lil-yc/aps360-project

Introduction

With the rise of generative music models (e.g. Suno, Udio, Riffusion), AI-generated music is becoming indistinguishable from human-composed music. This raises concerns around copyright, misinformation, and artist attribution. The goal of this project is to develop a deep learning model that can classify short music clips as either human-composed or AI-generated. Deep learning is suitable for this problem because it can learn to detect spectral artifacts or lack of long-term structural coherence that traditional heuristics might miss. CNNs and RNNs have demonstrated strong performance on audio classification tasks by learning hierarchical representations directly from time-frequency features. The primary model, a CNN, takes in Mel spectrograms (Figure 1) of AI or human music clips as an input, and outputs a 1 for AI-generated and 0 for human-composed. This report will discuss data processing, a baseline model, the primary CNN, and results.

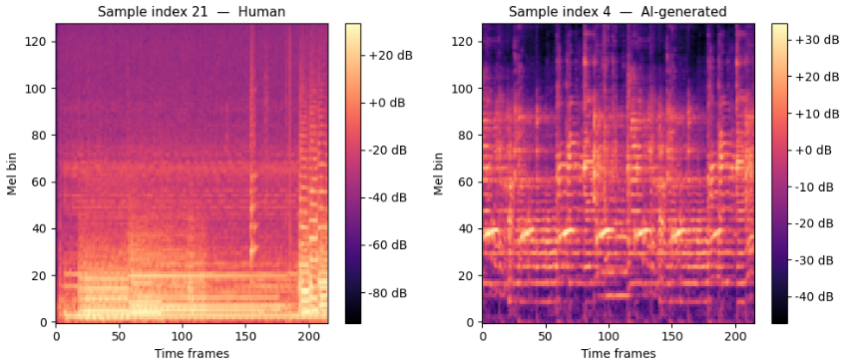

Figure 1: Mel spectrograms of a human vs. AI music clip.

Background & Related Work

Dieleman et al. [3] present one of the earliest works showing that CNNs trained on spectrograms can learn musically meaningful features without hand engineered descriptors. Li et al. [4] demonstrated that AI-generated audio exhibits detectable spectral artifacts; this concept is foundational in choosing deep learning as a solution. Demirörs et al. researched AI-generated speech detection [2], and determined that CNN models are the most suitable for detecting synthetic audio, compared to FC and SVM models. Afchar et al. [1] present one of the first general purpose AI music detectors, demonstrating that simple convolutional models can achieve over 99.8% accuracy on datasets containing real audio and artificial reconstructions. Wie et al. [5] also present a study on AI music detection, comparing several models with the highest accuracy being 99.7% achieved by a CNN. Based on these previous works, detecting AI-generated music using a CNN is a feasible and well-suited approach.

Data Processing

For the final model, 2000 Human-composed and AI-generated music tracks were collected from this dataset, with an even split of 1000 tracks per class. After loading the tracks, several forms of

audio processing were applied: The tracks were resampled at 22,050 Hz and converted to mono to ensure consistency. The tracks were cut down into 5 second clips so the resulting Mel spectrograms would be a consistent shape.

For the baseline model, Mel frequency cepstral coefficients were computed from each clip, and for the CNN, the clips were converted to Mel spectrograms (see Figure 1 for visualization).

For the baseline model, the data was split 80/20 for training and testing. There wasn't a need for a validation set since the baseline didn't need to be tuned. For the primary model, the data was split 70/15/15 for training, validation, and testing.

Since a separate test set (of 300 samples) was created for the primary model, this means that the model is being evaluated on new data. The test set was not used in training or hyperparameter tuning, so the test accuracies presented later are a good representation of the model's performance on new data.

Baseline Model

A baseline model was implemented by using the Mel frequency cepstral coefficients (MFCCs) of the audio clips to train a logistic regression classifier. After audio processing, 26 MFCCs were computed from each clip, so the logistic regression model takes only 26 features to fit to a curve. Since the model is a simple baseline, it did not require any tuning, and it was trained for 30 epochs.

As shown by the training loss curve for the baseline (Figure 2), the logistic regression model was able to converge and reach a training loss of 0.57 after 30 epochs, not lowering any further. The model was only able to achieve a test accuracy of 0.67, and average precision, recall, and f1-score of 0.67 as well. The model performs better than random guessing (50%) showing that there are detectable patterns, but the accuracy isn't very high, which shows the limitations of logistic regression and MFCCs for this task.

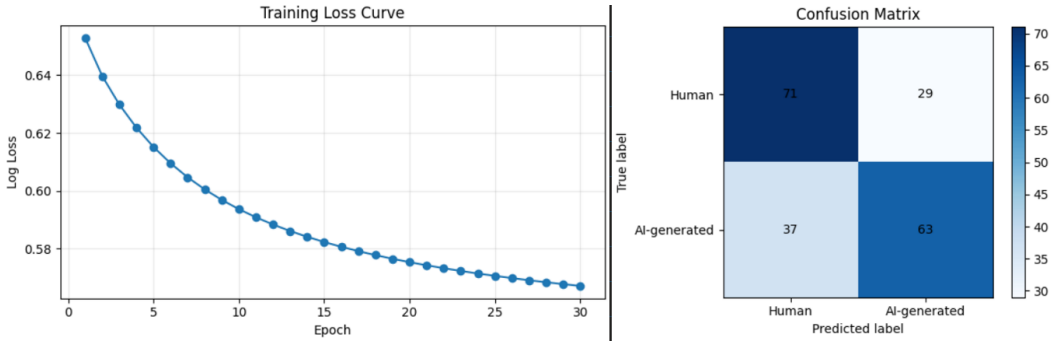

Figure 2: Training loss curve and confusion matrix of baseline model.

From the confusion matrix in Figure 2, there were 63 true positives, 71 true negatives, 29 false positives, and 37 false negatives. There seems to be a bias toward false negatives. A qualitative observation is that the false positives tend to have more processed and electronic sounds, and the false negatives tend to have more acoustic sounds, showing a potential bias with this model.

Architecture

The primary model (Figure 3) for this project is a CNN taking Mel spectrograms as inputs. The network consists of three convolutional blocks, with batch normalization and ReLU activations, followed by max pooling layers to progressively reduce dimensionality. The extracted features are passed through fully connected layers and a final sigmoid output unit for binary classification. Additionally, dropout regularization is applied, with dropout increasing after every layer to improve generalization. The loss function was binary cross entropy, and the Adam optimizer was used. The learning rate was 0.001, batch size was 16, and the model was trained for 30 epochs. Early stopping and model checkpointing was implemented in the training loop.

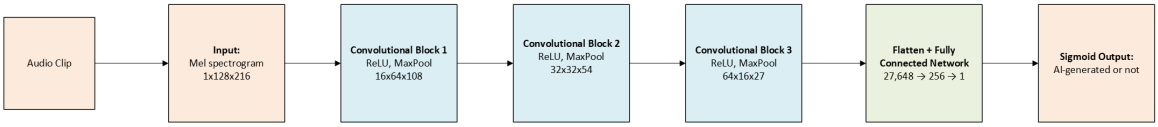

Figure 3: Diagram outlining CNN architecture.

Quantitative Results

The primary model was able to converge, as shown by the loss and error curves in Figure 4. It achieved a test accuracy of 0.970, precision 0.981, recall 0.963, and f1-score 0.972. The CNN has a significantly better performance compared to the logistic regression baseline, demonstrating that this architecture is suitable for the task.

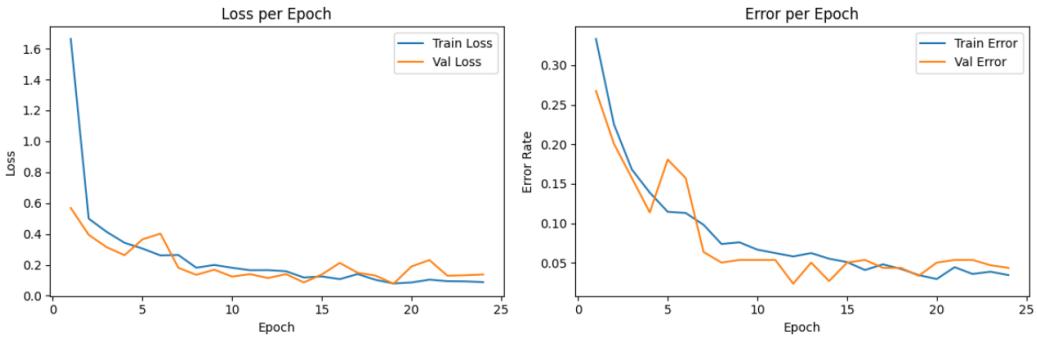

Figure 4: Test and validation loss and error curves of the primary model.

Qualitative Results

As shown by the confusion matrix (Figure 5) there are two times more false negatives than false positives. Mel spectrograms of sample inputs and outputs are included in Figure 6, though qualitative observations are more effectively made by listening to the audio samples: All 3 of the false positive were solo drum tracks, and the false negatives were all non-musical sound effects or ambience. This shows a potential difficulty in classifying non-melodic audio.

Discussion

Looking at the model's high test accuracy, 97%, as well as the other metrics previously discussed, the model is performing very well. From the qualitative observations, the model performs less accurately on audio that is less musical, which will be less of a problem since this model is specialized to classify music, not audio in general.

However, it's important to consider the source and size of the dataset in interpreting these results. All of the human-composed music comes from one source which collects royalty-free music, and the AI-generated music was generated by a few of the same models, which limits the variety of music in the dataset. Since the total data used was only 2000 samples, the test metrics were computed on only 300 samples. Taking this into account, the model may not generalize well in real-world conditions.

While having a larger dataset would have trained and tested the model more accurately, downloading full audio tracks was very time-consuming and takes far longer compared to downloading images. and the Mel spectrogram conversion provides an additional overhead. Additionally, AI-music detection is a relatively newer problem, so there is less data available.

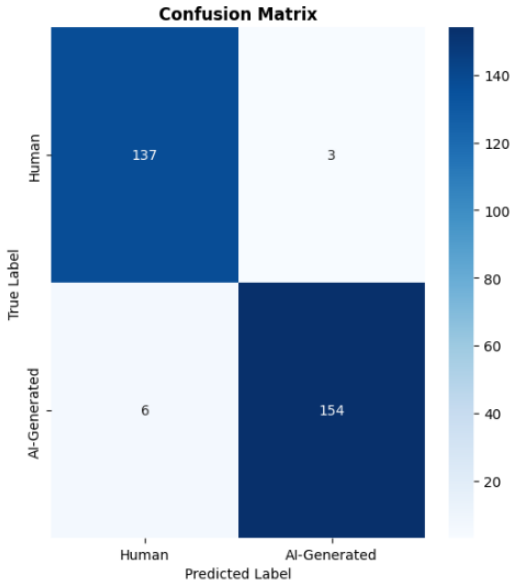

Figure 5: Confusion matrix of the primary model.

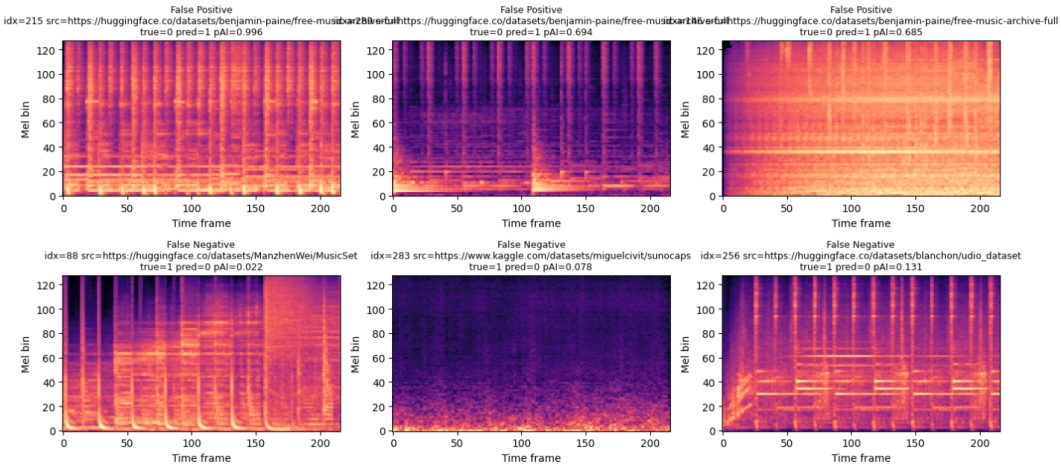

Figure 6: Mel spectrograms of false positives and false negatives

Ethical Considerations

A major ethical concern in this project is the risk of false positives, where human-composed music is incorrectly classified as AI-generated. False negatives are also a concern, allowing AI-generated music to pass screening. These errors could negatively impact artists and consumers if used in real world moderation or copyright enforcement systems. Dataset bias toward certain genres or production styles may also affect model fairness. Considering these limitations, this model should not be used to make significant judgments and is developed for learning purposes.

References

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[2] Meriç Demirörs, Toygar Akgün, and Ahmet Murat Özbayoğlu. Ai generated speech detection using cnn. In *2024 IEEE International Conference on Big Data (BigData)*, pages 7002–7010, 2024.

[3] Sander Dieleman and Benjamin Schrauwen. End-to-end learning for music audio. In *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 6964–6968, 2014.

[4] Menglu Li, Yasaman Ahmadiadli, and Xiao-Ping Zhang. A survey on speech deepfake detection. *ACM Computing Surveys*, 57(7):1–38, February 2025.

[5] Jason Wie, Nicholas Salim, Alexander Agung Santoso Gunawan, and Rilo Chandra Pradana. Detecting ai-generated music: A comparative analysis of deep learning models. In *2025 Tenth International Conference on Informatics and Computing (ICIC)*, pages 1–6, 2025.